Water Retention Curve Fitting Tool

User Manual

Version 1.0

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1. Introduction

The Water Retention Curve Fitting Tool is designed to estimate hydraulic parameters from experimental water retention data using a modified van Genuchten model.

The software supports both mineral soils and horticultural growing media such as peat, wood fiber substrates, compost-based media, and substrate mixtures.

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2. Installation

1. Download:

WaterRetentionCurveFitting_Setup.exe

2. Run the installer.
3. Follow the installation wizard.
4. Launch the software from the desktop or Start Menu.

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3. Input Data

The software requires two columns of data:

- Pressure Head (hPa)

- Volumetric Water Content (θ)

Example:

Pressure Head (hPa) Water Content

10 ---- 0.50

30 ---- 0.48

100 ---- 0.43

300 ---- 0.32

1000 ---- 0.24

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4. Loading Data

Click:

Load Excel

or

Load Data

and select the input file.

The imported values will appear in the data table.

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5. Initial Guess

Open the Initial Guess tab.

Provide initial estimates for:

- α
- n
- m
- θ_r
- θ_s

- k

Reasonable initial values generally improve convergence.

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6. Parameter Bounds

Open the Bounds tab.

Specify lower and upper limits for each parameter.

Bounds help ensure physically meaningful solutions.

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7. Curve Fitting

Click:

Fit Curve

The software will:

- Optimize model parameters
- Plot measured data
- Plot fitted curve
- Display goodness-of-fit statistics

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8. Exporting Figures

Click:

Save Figure (600 DPI)

Supported formats:

- PNG
- TIFF
- JPEG
- PDF
- SVG

The exported figures are suitable for scientific publications.

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9. Modified van Genuchten Model

The software includes an empirical parameter k.

The parameter k is introduced to improve fitting performance for horticultural substrates and organic growing media.

It should not be interpreted as a physical hydraulic parameter.

Instead, it acts as a tuning parameter that allows the model to better represent water retention behavior when residual water content remains above zero.

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10. Citation

If you use this software in research, please cite:

Muhammed, H.H. and Anlauf, R.

Water Retention Curve Fitting Tool (Version 1.0). ©2026

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11. Contact

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